

DEPRESSION DETECTION PLATFORM - COMPLETE PROJECT EXPLANATION

Project Phase II (CS421) - Department of Computer Science & Engineering

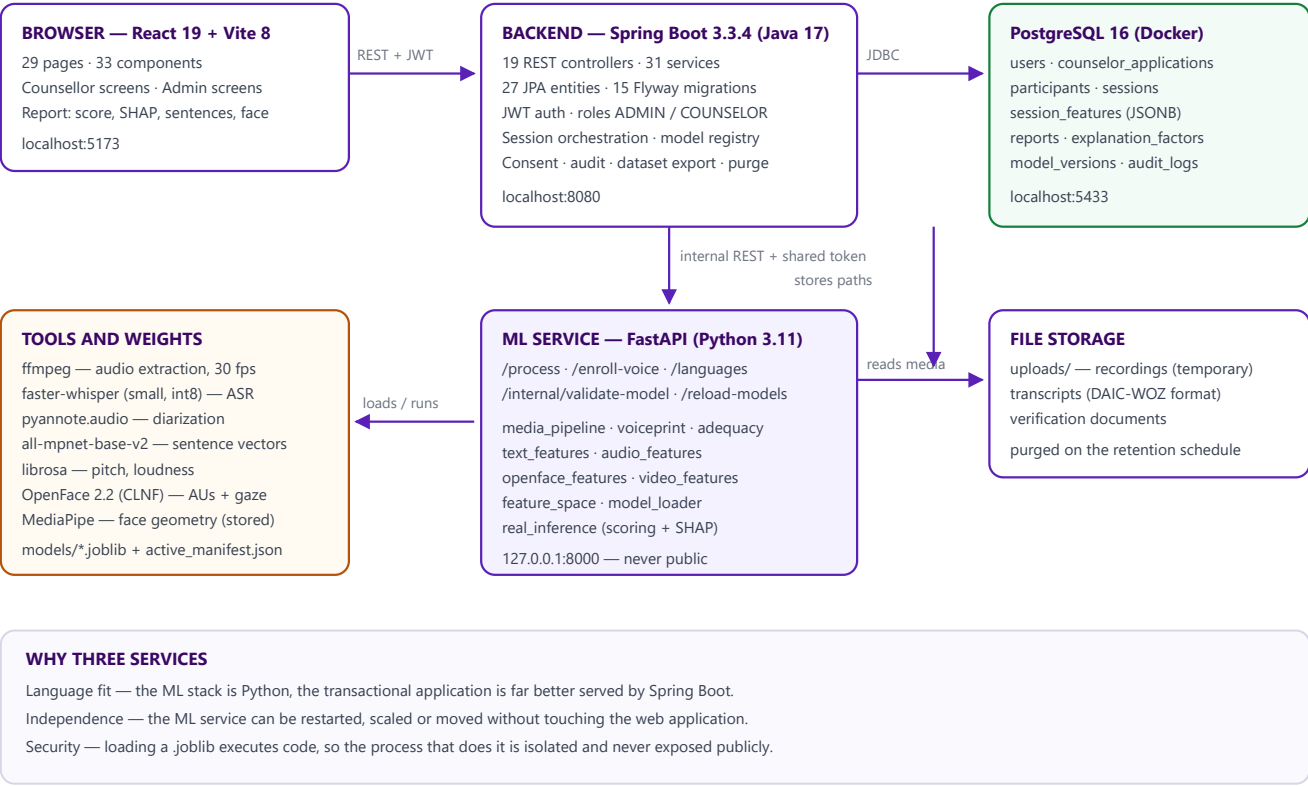
Prepared 12 September 2026. Every number in this document was measured, not estimated.

HOW TO USE THIS FILE

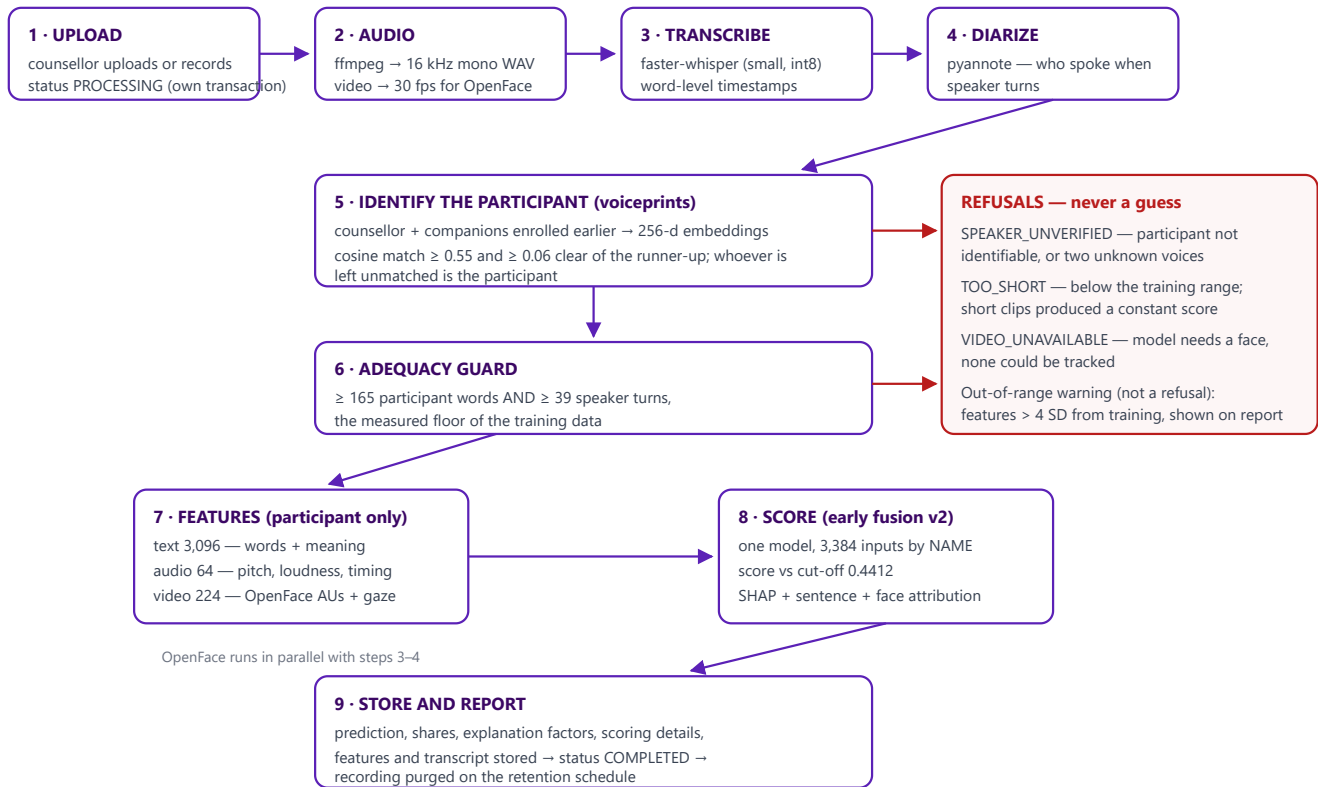
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A. ARCHITECTURE AND IMPLEMENTATION DIAGRAMS

Five diagrams of the system as it actually runs today: the services, one session end to end, the inside of the shipped model, how a prediction is explained, and how a model reaches production.

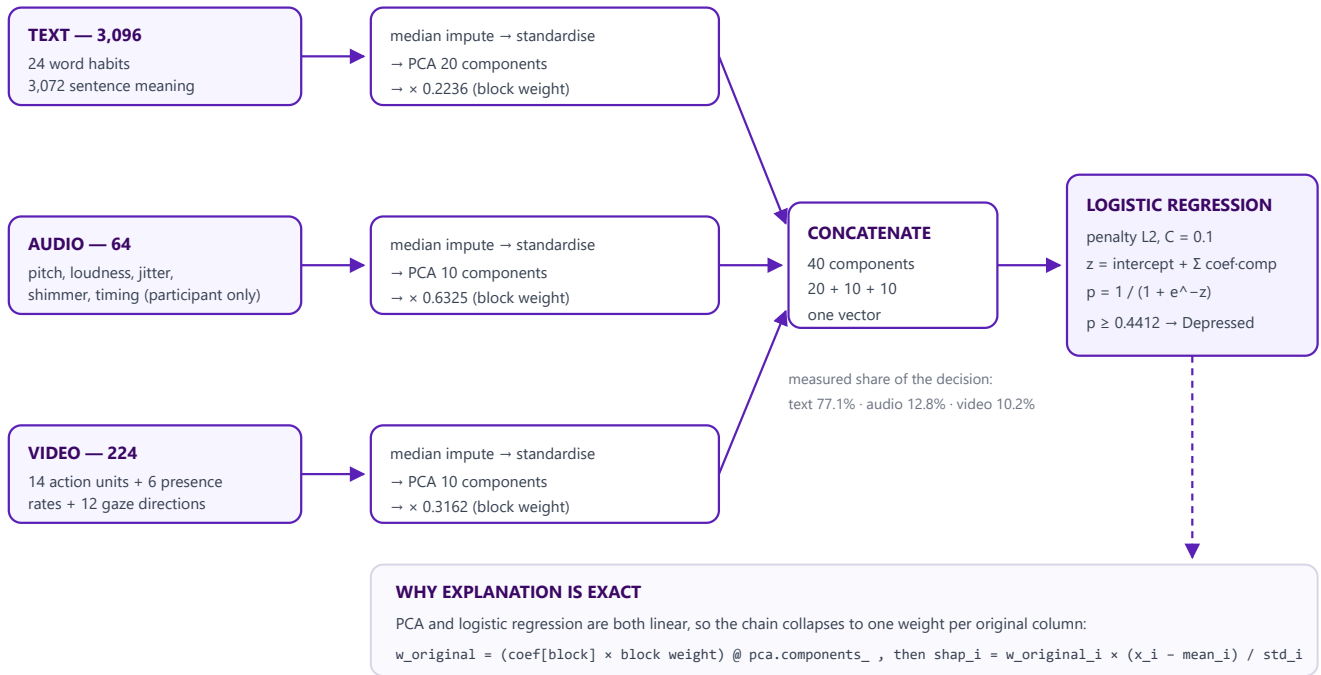


A1 — System architecture. Three services, one database, one file store. The frontend never talks to the ML service directly; the backend is the only caller.

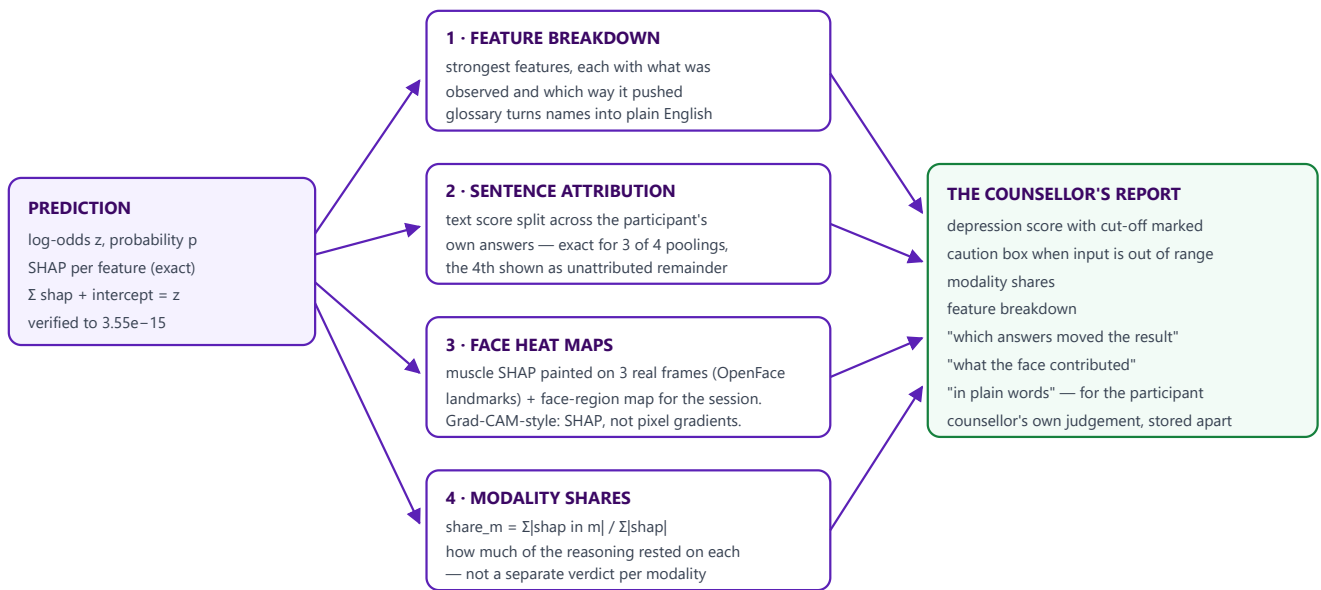


A2 — One session, end to end. Steps 3 and 4 run while OpenFace tracks the face in a separate process, so the video analysis costs almost no extra wall-clock time.

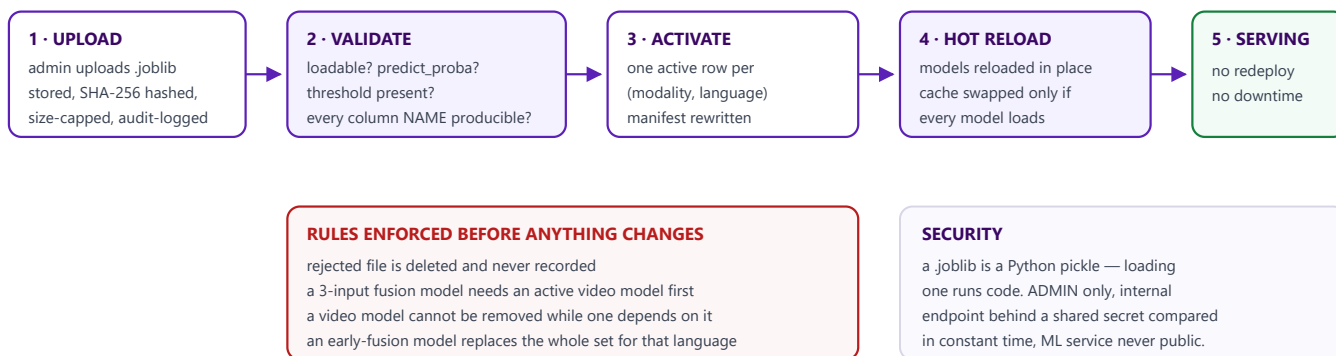
EARLY FUSION v2 — one model on all 3,384 measurements



A3 — Inside the shipped model. Per-block PCA plus block weights are what stop the 3,096 text columns from swamping 64 audio and 224 video columns.



A4 — One prediction, four kinds of explanation. Everything shown on a report is derived from the same exact SHAP decomposition, never from a separate explainer model.



A5 — How a retrained model reaches production without a developer: upload, validate by column name, activate, hot reload.

1 PROBLEM STATEMENT

Depression is common, treatable, and badly under-detected. In a first counselling conversation the counsellor has only their own impression to work from: there is no instrument that measures what was said and how it was said, and no record of the evidence behind that impression. Self-report questionnaires (PHQ-8/PHQ-9) depend on the person answering accurately in the moment, and are easy to under-answer.

Formal problem statement:

"Given an audio-video recording of a counselling conversation, produce an evidence-backed screening indicator of depression for the participant only, which (a) separates the participant's speech from everyone else's, (b) measures what they said, how they said it, and how their face moved, (c) combines those measurements into a single score with a stated decision cut-off, (d) explains that score feature by feature in language a non-technical reader can act on, and (e) refuses to answer when the recording is outside the range the models were trained on - all inside a secured, consent-governed web platform that a counsellor and an administrator can operate without a developer."

The last two clauses are what separate this from a notebook experiment: an unexplained number is not usable in mental health, and a confident number produced from unusable input is actively dangerous.

2 OBJECTIVES, AND WHAT WAS ACHIEVED

O1. Build a multimodal ML pipeline (text + audio + video) on a recognised clinical

corpus.
ACHIEVED - DAIC-WOZ, 188 participants.

O2. Train honestly: official splits, one untouched test set, repeated validation.

ACHIEVED - 10 repeated 80/20 splits with the whole model selection redone inside every split, plus a single evaluation on the official test split.

O3. Beat the previously available baseline on the same protocol.

ACHIEVED - AUC 0.8463 vs 0.768, F1 0.6857 vs 0.611, accuracy 0.766 vs 0.702.

O4. Make every modality genuinely contribute to the decision.

ACHIEVED in v2 - text 77.1%, audio 12.8%, video 10.2% of the decision weight, with the accuracy cost measured and reported (-0.0577 cross-validated AUC).

O5. Explainable AI that a participant can understand.

ACHIEVED - exact SHAP values (verified to 3.55e-15), per-sentence attribution, per-face-region attribution, and a plain-language section on every report.

O6. Deployable by an operator, not a developer.

ACHIEVED - upload, validate and activate models from the admin screen, with hot reload and no redeploy.

O7. Consent, privacy and auditability by design.

ACHIEVED - consent record per session, recordings purged after feature extraction, per-participant erasure, full audit log.

O8. Refuse rather than guess.

ACHIEVED - TOO_SHORT, SPEAKER_UNVERIFIED and VIDEO_UNAVAILABLE refusals, plus out-of-range warnings printed on the report itself.

3 THE DATASET: DAIC-WOZ

3.1 What it is

DAIC-WOZ = Distress Analysis Interview Corpus - Wizard of Oz, University of Southern California (Gratch et al., LREC 2014). Clinical-style interviews conducted by an on-screen virtual interviewer called "Ellie", operated by a human in another room (the "wizard"). It is the benchmark corpus of the AVEC 2016 and AVEC 2017 depression sub-challenges. Access requires an approved request; ours was approved. The corpus may NOT be redistributed, which is why it is kept out of the git repository.

3.2 What each participant folder contains

- AUDIO.wav	16 kHz mono 16-bit PCM, whole session
- TRANSCRIPT.csv	start_time, stop_time, speaker, value (one row per utterance)
- COVAREP.csv / FORMANT.csv	pre-computed audio features at 100 Hz (not used by us)
- CLNF_features/_features3D	68 facial landmarks per frame (30 fps)
- CLNF_AUs.txt	facial action-unit intensities and presence flags
- CLNF_gaze.txt	gaze direction, in camera frame AND head frame
- CLNF_pose.txt	head translation and rotation
- CLNF_hog.bin	appearance descriptors (not used by us)

NOTE: DAIC-WOZ ships NO raw video. This single fact decides several design choices later in this document - most importantly, true Grad-CAM is impossible, and section

18.4 explains the Grad-CAM-style heat map drawn instead.

3.3 Cohort actually used (measured in our own Phase-1 audit)

- Participants listed in the official split files	189
- Folders present on disk	188
- Excluded: participant 440	corrupt in the official release
- Release-documented exclusions (already absent)	342, 394, 398, 460
- USED	188
- Depressed 55 (29.3%) / Not depressed 133 (70.7%)	
- PHQ-8 score: min 0, median 5, mean 6.68, max 23	
- One label anomaly: one participant has PHQ8_Score = 10 but PHQ8_Binary = 0. We kept the released label rather than "fixing" it, and reported the anomaly.	

3.4 Participant speech volume (annotation tokens stripped) - this defines our guards

	min	5th pct	median	max
utterance rows	42	76	150.5	385
words	165	489	1247.5	4551
duration (minutes)	6.9	10.0	15.2	32.8

Three transcripts contain no interviewer rows at all.

3.5 LABELLING - two different labels, never mixed

(a) TRAINING LABEL (DAIC-WOZ): PHQ8_Binary. The 8-item Patient Health Questionnaire

total (0-24) thresholded at ≥ 10 , the standard cut-off for probable depression.
It is a questionnaire outcome, not a psychiatrist's diagnosis - an important honesty point in the viva.

(b) PLATFORM LABEL (our own research dataset): the COUNSELLOR'S OWN ASSESSMENT,

recorded on the report. The model's prediction is NEVER used as a label, because training on your own output teaches nothing and launders error into "truth".

The two live in separate database tables and are never averaged together.

3.6 Official split

train 107 - dev 34 - test 47 participants, participant-disjoint (nobody appears in two splits). Class balance is similar across splits (about 28-30% depressed).

4 WHY THIS CORPUS IS HARDER THAN THE LITERATURE SUGGESTS

Burdisso et al. (ClinicalNLP @ NAACL 2024) showed that a model reading ONLY the interviewer's prompts reaches about 0.90 F1 on DAIC-WOZ - the highest text-only result reported on the corpus - because the interview branches on the participant's answers, so which questions Ellie asked leaks the outcome. Any model trained on the full transcript is partly exploiting that, and the leak disappears in real deployment.

Rules we adopted as a result, all of them costly:

- Train on the PARTICIPANT'S SPEECH ONLY. Interviewer turns are dropped everywhere.
- Remove interviewer-dependent audio timing. 24 of the original 85 audio features measured response latency and between-turn gaps - that is, Ellie's scripted pacing. Removing them costs accuracy (audio-only cross-validated AUC 0.671 -> 0.609) but what remains is genuinely the participant's voice.
- Never use gender. The training split has an 18-point depression-rate gap between genders that does not exist in dev or test: a shortcut that would inflate our numbers and then fail on real people.

This is why our numbers look lower than some published ones, and why we can defend them.

5 FEATURES: WHAT THE COMPUTER ACTUALLY MEASURES

5.1 TEXT - 3,096 numbers per session

(a) 24 lexical features, counted from the participant's cleaned words:

total words; number of answers; words per answer (mean/std/median); type-token ratio; root type-token ratio; mean word length; rates of first-person singular ("I, me, my"), first-person plural, second person, third person; absolutist words ("always, never, nothing"); negative-emotion words; positive-emotion words; negations; hedges ("maybe", "I guess"); past-tense words; sleep/fatigue words; self-focus ratio; negative-to-positive ratio; sentiment balance; very-short-answer rate; long-answer rate.
These are research-backed markers (self-focus, absolutist language, negative affect) and, crucially, a human can read them.

(b) 3,072 sentence-meaning features: every answer is embedded with

sentence-transformers/all-mpnet-base-v2 (768 dimensions per sentence, maximum 64 tokens, float32), then pooled four ways and concatenated:
[mean | length-weighted mean | max | standard deviation] x 768 = 3,072
The max block matters clinically: one very dark sentence in an otherwise flat interview is exactly what an average would erase.

Block order is fixed and is part of the model contract. Getting it wrong produces a confident, wrong answer that nothing downstream can detect.

5.2 AUDIO - 64 numbers per session (the "clean-64" set)

Computed with librosa over participant speech only:

- utterance duration and words-per-utterance, each summarised by 9 functionals (mean, std, p10, p25, p50, p75, p90, inter-quartile range, slope)
- total speech time, number of utterances, total words, words per second
- loudness: RMS in dB, 9 functionals plus range
- voiced ratio
- pitch: YIN estimator, 65-400 Hz, converted to semitones, 9 functionals plus range
- jitter proxy (frame-to-frame pitch change) and shimmer proxy (frame-to-frame loudness change), 9 functionals each
- three participant-only timing features re-derived so they never reference the interviewer: session_len_s_pt, speech_ratio_pt, utts_per_minute_pt

Clinical grounding: flattened pitch, reduced loudness variation and slowed speech are documented correlates of depression (psychomotor retardation).

5.3 VIDEO - 224 numbers per session (OpenFace)

From OpenFace 2.2 using the CLNF landmark model, the same family DAIC-WOZ used:

- 14 action-unit intensities (AU01, 02, 04, 05, 06, 09, 10, 12, 14, 15, 17, 20, 25, 26), each summarised by 8 statistics - mean, std, p10, p50, p90, max, and two movement statistics (mean and std of frame-to-frame change) = 112
 - 6 action-unit presence rates (AU04, 12, 15, 23, 28, 45) = 6
 - au_total, the sum of the 14 intensities, with the same 8 statistics = 8
 - gaze: 12 columns (both eyes; camera-frame x/y/z and head-frame x/y/z), 8 statistics each = 96
 - au_frac_tracked and gz_frac_tracked, how much of the session was trackable = 2
- TOTAL = 224

Frames are filtered to success == 1 and confidence >= 0.75, and every recording is resampled to 30 fps first, because the movement statistics scale with frame rate. Separately, the platform also stores 111 MediaPipe face-geometry features for its own future research dataset; those are not used by the DAIC-WOZ models.

What an Action Unit is: the Facial Action Coding System describes facial movement as individual muscle actions. AU04 is the brow lowerer (frowning), AU12 the lip-corner puller (smiling), AU15 the lip-corner depressor, AU17 the chin raiser, AU45 a blink.

6 FROM RECORDING TO FEATURES: THE INFERENCE PIPELINE

1. The counsellor uploads a session recording (common formats, up to 520 MB). 2. ffmpeg extracts 16 kHz mono audio. 3. faster-whisper (model "small", int8 quantised, CPU) transcribes with word-level

timestamps.

4. pyannote.audio (speaker-diarization-community-1) decides who spoke when. 5. SPEAKER IDENTIFICATION by voiceprint. The counsellor - and any companion in the

room - enrolled earlier by reading a fixed passage, giving a 256-dimensional voice embedding. Each diarized speaker is compared to those by cosine similarity. A speaker is claimed by a known voice if similarity ≥ 0.55 AND it beats the runner-up by ≥ 0.06 . Whoever is left unclaimed is the participant. If that is ambiguous - no one left over, or more than one unknown voice - the session is REFUSED (SPEAKER_UNVERIFIED) rather than scoring the wrong person.

6. ADEQUACY GUARD. If the participant produced fewer than 165 words or fewer than 39

speaker turns, the session is refused as TOO_SHORT. Section 7.5 explains the real bug that made this necessary.

7. FEATURE EXTRACTION, participant speech only: text 3,096; audio 64; video 224 via

OpenFace, which runs as a separate process IN PARALLEL with transcription so it adds almost nothing to the wall-clock time.

8. Each model is fed BY COLUMN NAME from its own stored list of column names, so a

model can never silently receive the wrong feature set.

9. Scoring (section 7), explanation (section 18), report stored, and the recording

purged on the retention schedule.

7 THE MODELS, AND HOW THEY WERE TRAINED

Three generations of model exist in this project. All of them are trained on the same 188 DAIC-WOZ participants with the same rules; they differ in how the modalities are combined.

7.1 The building block: one scikit-learn Pipeline per modality (LATE FUSION)

```
SimpleImputer(median) -> StandardScaler -> SelectKBest(ANOVA F) -> LogisticRegression
```

- SimpleImputer(median): a feature that could not be measured arrives as NaN (never as 0, which would be a real measurement of "none") and is filled with the median of the training data.
- StandardScaler: each feature becomes $(x - \text{mean}) / \text{std}$, so features with different units are comparable.
- SelectKBest: keeps the k features most associated with the label by ANOVA F-test. With 188 people and 3,096 text features, selection is not optional - without it a model memorises the training split.
- LogisticRegression: a linear model, chosen deliberately so every prediction can be explained exactly (section 18).

7.2 GENERATION 1 - LATE FUSION (three models plus a combiner)

```
text : SelectKBest k=50, C=0.01 -> test AUC 0.7532
audio : SelectKBest k=25, C=1.0 -> test AUC 0.6017
video : SelectKBest k=25, C=1.0 -> test AUC 0.5844
fusion: LogisticRegression over the three probabilities [p_text, p_audio, p_video]
        trained on OUT-OF-FOLD probabilities (see 7.6)
        coefficients: text +1.988, audio +0.909, video -0.510, intercept -1.633
        threshold 0.2945
        test: AUC 0.7684, accuracy 0.7021, F1 0.6111, sensitivity 0.786, specificity 0.667
```

Reading of those coefficients: the combiner learned to trust text most, audio less, and to treat the video probability as mildly NEGATIVE evidence - on this corpus the video model was at or below chance, so the combiner used it as a correction, not as support.

7.3 GENERATION 2 - EARLY FUSION v1 (one model on everything, joined)

```
All 3,384 features (3,096 text + 64 audio + 224 video) are concatenated into ONE
input vector, and one model is trained on it:
ColumnTransformer(
    text    -> Imputer -> Scaler -> PCA(40)
    audio64 -> Imputer -> Scaler -> PCA(10)
    video   -> Imputer -> Scaler -> PCA(10) ) -> LogisticRegression(elastic-net)
```

Why PCA per block rather than one SelectKBest over all 3,384 columns: we tested the naive version and it reproduced the classic early-fusion failure exactly - a global SelectKBest chose 50 text columns and ZERO audio or video columns, so "early fusion" returned byte-identical numbers to a text-only model. That is demonstrated in our results, not asserted.

Outcome: official-test AUC 0.8333, accuracy 0.766, F1 0.667 - but the elastic-net penalty (which can drive coefficients to exactly zero) had deleted the ENTIRE audio block. Weight shares came out text 97-98%, audio 0.0%, video 2-3%. It was multimodal in name only.

7.4 GENERATION 3 - EARLY FUSION v2 (the model this project ships)

Two changes fix the previous problem:

- (a) L2 penalty only. L2 shrinks coefficients but never sets them exactly to zero, so no block can be deleted. (L1 / elastic-net is what deleted audio in v1.)
- (b) Block balancing. Each block's PCA output is multiplied by a weight $w_b / \sqrt{k_b}$ using `ColumnTransformer(transformer_weights=...)`. Without this, the text block dominates by construction simply because it has more columns and more components. The weights were tuned inside cross-validation.

Final shipped configuration:

```
text      3,096 columns -> PCA 20 components, block weight 0.2236
audio64    64 columns -> PCA 10 components, block weight 0.6325
video     224 columns -> PCA 10 components, block weight 0.3162
classifier: LogisticRegression(penalty='l2', C=0.1) over the 40 components
decision threshold 0.4412 (0.3449 when refitted under the official protocol)
```

Acceptance rule - stated openly as a PRODUCT REQUIREMENT, not a data finding:

text share must be the largest, audio $\geq 10\%$, video $\geq 5\%$ of the back-mapped coefficient mass. 90 of the 384 candidate configurations met it; the best of those by cross-validated AUC was shipped.

Achieved shares: text 77.1%, audio 12.8%, video 10.2%.

THE COST, measured and reported rather than hidden:

```
cross-validated AUC 0.7633 (unconstrained) -> 0.7056 (constrained) = -0.0577
seed-42 test AUC      0.8249 -> 0.7946
```

In other words: forcing audio and video to count makes the model slightly worse on this corpus. We ship the constrained model because the project requires a genuinely multimodal decision, and we say so.

7.5 THE DECISION THRESHOLD, and the bug that taught us to guard the input

A classifier outputs a probability; a threshold turns it into a decision. We do NOT use 0.5, because only ~30% of the training participants are depressed - at 0.5 the model would miss most of them. The threshold is chosen by YOUDEN'S J STATISTIC:

$J = \text{sensitivity} + \text{specificity} - 1$, maximised over candidate thresholds, computed on OUT-OF-FOLD predictions of the training data only - never on test.

The bug that this project is most careful about: in an early version, every SHORT recording came back "depressed". The cause was not the threshold but the input: a one-minute recording is far outside the training range, the text model's RBF kernel collapsed to a constant 0.2542 for every such session, and 0.2542 happened to sit just above the then-threshold of 0.48 after fusion. A model fed data unlike its training data still returns a confident number. Hence the adequacy guard (165 words / 39 turns) and the out-of-range warnings on every report.

7.6 EVALUATION PROTOCOL (the part that makes the numbers trustworthy)

- PRIMARY: 10 repeated stratified 80/20 participant splits. Inside each 80%, 5-fold cross-validation chooses PCA sizes, block weights, C and the threshold; the 20% is scored once. Crucially the WHOLE selection is redone inside every split, so the mean is not inflated by choosing a winner on one lucky split. We learned this the hard way: on a single split, early fusion looked like a clear winner (0.7736 vs 0.7343); across 10 splits that reversed.
- SECONDARY: the official AVEC-2017 protocol - train on train+dev (141 people), score the official 47-person test split ONCE. This is the number comparable with published work.
- Confidence intervals: 10,000-sample bootstrap over test participants.
- Out-of-fold probabilities: for the late-fusion combiner, the three model probabilities are produced by `cross_val_predict` so the combiner never sees predictions made on data the base models trained on. Otherwise it would learn from over-confident inputs and set the wrong weights.

8 EARLY FUSION vs LATE FUSION (the concept your mentor will ask about)

LATE FUSION (decision-level): train one model per modality, then combine their OUTPUT PROBABILITIES.

- + Each modality can use the algorithm that suits it; missing modalities are easy to handle; the combiner's weights are directly interpretable.
- The modalities never interact: any relationship between "flat voice" AND "negative words" is invisible, because each model has already collapsed to one number.

EARLY FUSION (feature-level): concatenate all features into ONE vector and train ONE model on it.

- + The model can use relationships ACROSS modalities, because it sees all the raw measurements together.
- Scale and width imbalance destroys it if handled naively: 3,096 text columns next to 64 audio columns means the selection or the regularisation simply throws the small blocks away. That is exactly what we measured in v1.

Our solution - the standard one in the literature and the one we implemented:

per-block dimensionality reduction (PCA) so each modality contributes a comparable number of components, plus per-block weights so no modality dominates by width, and an L2 penalty so no block can be zeroed.

INTERMEDIATE / HYBRID FUSION exists too (e.g. gated or attention-based fusion at utterance level). The earlier research this project inherited tried six such architectures - BiLSTM, transformer, cross-attention, gated and bilinear fusion - and all of them lost to simple models on this corpus, because 107-142 training people is far too few to fit them honestly.

9 THE ARITHMETIC, END TO END (worked, with real numbers)

Everything below is the shipped v2 early-fusion model.

STEP 1 - Impute. Any unmeasured feature becomes the training median.

STEP 2 - Standardise. For each column i:

```
z_i = (x_i - mean_i) / std_i
mean_i and std_i are the training values, stored in the model bundle
(feature_means, feature_stds). Note: for a column with zero variance,
StandardScaler stores scale_ = 1.0, NOT 0.0 - dividing by the raw std there
would be division by zero. (The GPU run caught and fixed exactly this.)
```

STEP 3 - PCA per block. For block b with k_b components:

```
c_b = (z_b - pca_mean_b) @ components_b^T          (shape: k_b numbers)
text -> 20 components, audio -> 10, video -> 10. Total 40.
```

STEP 4 - Block weighting. Each block's components are multiplied by its weight:

```
text x 0.2236, audio x 0.6325, video x 0.3162
(each weight is w_b / sqrt(k_b), so wide blocks do not win by width alone).
```

STEP 5 - Logistic regression over the 40 weighted components:

```
z (the log-odds) = intercept + sum_j coef_j * component_j
intercept = -1.2749 in the shipped model.
```

STEP 6 - Probability (the percentage on the report):

```
p = 1 / (1 + e^(-z))          [the sigmoid / logistic function]
```

STEP 7 - Decision:

```
p >= threshold (0.4412) -> "Depressed"
p < threshold          -> "Not depressed"
```

WORKED EXAMPLE (verification participant 458, from the GPU run):

```
sum of all SHAP values + intercept = z = -6.7561
p = 1 / (1 + e^(6.7561)) = 0.00116 = 0.1%
0.001 < 0.4412 -> Not depressed. True label: Not depressed. Correct.
```

SECOND EXAMPLE (participant 384):

```
z = +0.2713 -> p = 1 / (1 + e^(-0.2713)) = 0.5674 = 56.7%
0.567 >= 0.4412 -> Depressed. True label: Depressed. Correct.
```

BACK-MAPPING A COEFFICIENT TO ONE ORIGINAL FEATURE (this is what makes explanation possible at all). PCA is a linear map, so the chain scaler -> PCA -> linear model is itself linear in the original columns:

```
w_original = (coef[block's components] * block_weight) @ components_b
giving one weight per original column. The block weight MUST be included; if it
is left out every contribution and every modality share is silently wrong. (We
found and fixed exactly this in the website's code when v2 arrived.)
```

SHAP VALUE OF FEATURE i (exact, no sampling):

```
shap_i = w_original_i * (x_i - mean_i) / std_i
and the additivity property holds exactly:
sum_i shap_i + intercept = z = the model's log-odds
Verified on 5 held-out participants: maximum difference 3.553e-15, which is
floating-point noise. Cross-checked against the shap library's LinearExplainer:
maximum per-feature difference 6.193e-16.
```

MODALITY SHARE (what the report shows as "text 77%, audio 13%, video 10%"):

```
share_m = sum over features in m of |shap_i| / sum over ALL features of |shap_i|
It answers "how much of the model's reasoning rested on this modality", which is
NOT the same as "this modality's own verdict" - a single early-fusion model has
no per-modality verdict, and the report says so.
```

THRESHOLD SELECTION (Youden's J):

```
J(t) = sensitivity(t) + specificity(t) - 1, maximised over t,
computed on out-of-fold training predictions only.
```

CONFUSION MATRIX AND THE METRICS (official protocol, shipped v2 model):

```
TN = 24    FP = 9
FN = 2     TP = 12
accuracy   = (TP+TN)/total = 36/47 = 0.766
sensitivity = TP/(TP+FN)    = 12/14 = 0.857    (depressed people found)
specificity = TN/(TN+FP)    = 24/33 = 0.727    (non-depressed cleared)
precision   = TP/(TP+FP)    = 12/21 = 0.571
F1          = 2*P*R/(P+R)   = 0.6857
balanced accuracy = (0.857+0.727)/2 = 0.792
ROC-AUC     = 0.8463
```

10 WHY THE OUTPUT IS A PERCENTAGE, AND WHAT IT IS NOT

The number on the report is the model's DEPRESSION SCORE: the probability it assigns to the "depressed" class, from the sigmoid in step 6. Three things to be clear about:

- It is NOT a severity. 80% does not mean "more depressed" than 60%; it means the pattern resembles the depressed group more strongly.
- It is NOT the model's confidence in its own correctness.
- It is compared against a CUT-OFF THAT IS NOT 50%. For the shipped model the cut-off is 44% (and 34% under the official-protocol refit; the v1 late-fusion model used 29%). A score of 44% therefore means "Depressed". This confused a real reader during development - a report read "Depressed, 35% confidence" - which is why the report now labels the number "depression score", draws the cut-off as a tick on the ring, and explains in one sentence why the cut-off is below half.

Why the cut-off is below 50%: only about 3 in 10 people in the training data were depressed, so the model's probabilities sit low overall; a 0.5 cut-off would miss most depressed participants. The threshold is set where sensitivity and specificity are jointly best (Youden's J).

11 GUARDS: WHEN THE SYSTEM REFUSES TO ANSWER

TOO_SHORT	participant produced < 165 words or < 39 speaker turns. Measured floors from the training split. A shorter interview is outside the range every model was fitted on, and the failure mode is not "slightly worse" but "confidently wrong in one direction".
SPEAKER_UNVERIFIED	the participant could not be identified unambiguously - the counsellor's voice was not found, nobody was left over, or more than one unknown voice was present.
VIDEO_UNAVAILABLE	the serving model needs facial measurements and no face could be tracked. Substituting a neutral value would silently change the arithmetic, so the session is refused instead.
FAILED	anything else (ML service unreachable, unreadable media), with the reason stored on the session.

OUT-OF-RANGE WARNING (not a refusal) every session also reports how many of the

	features the model uses sit more than 4 standard deviations from the training data, and names the worst one. This is printed on the report so a counsellor knows when to discount the score. Typical causes: microphone level, camera and lighting, interview length.
--	---

12 SYSTEM ARCHITECTURE

Three services, each doing what it is good at:

BROWSER (React 19 + Vite)

```
    | REST + JWT (Bearer token)
    v
BACKEND (Spring Boot 3.3.4, Java 17) <---- JDBC ----> PostgreSQL 16 (Docker, port 5433)
    | internal REST, shared-secret token
    v
ML SERVICE (Python 3.11, FastAPI) ----> model files (.joblib), ffmpeg, Whisper,
                                         pyannote, OpenFace, MediaPipe
```

Why three services rather than one:

- The ML ecosystem is Python; the business application (security, transactions, relational data) is far better served by Spring Boot. Forcing either into the other's language would be a permanent tax.
- The ML service can be restarted, scaled or kept OFF the public internet independently. It is the component that loads .joblib files, and loading a .joblib executes code, so isolating it is a security decision as much as an engineering one.
- Models can be replaced without touching or redeploying the Java code.

12.1 HOW THE DESIGN WAS ARRIVED AT (the process, not just the result)

1. Contract first. The ML service's response shape was fixed before any real model

existed, and a mock pipeline returned that exact shape. That is why swapping in real models, then a three-input fusion model, then an early-fusion model never required a change to the backend or the frontend contract.

2. Database before features. Sessions, consent and reports were modelled first; feature

vectors were added later as JSONB precisely so a model with a different width would never force a schema migration.

3. Refusals designed in, not bolted on. Each guard (too short, unidentified speaker,

missing face) was added the moment the failure was understood, and each one returns a status the counsellor can act on rather than a stack trace.

4. One stylesheet, tokens not components. A single design system (violet accent, warm

neutral ground, red reserved for real errors) with reusable loading/empty/error states, so 29 pages look like one product and a new page inherits the behaviour.

5. Explanation treated as a feature, not a report footer. When the text model became

linear, the report gained per-feature reasons; when sentence vectors became available, it gained per-sentence reasons; when OpenFace arrived, per-region ones.

6. Every change verified against the corpus rather than assumed - which is how the gaze

bug and the constant-output bug were both caught.

Size of the system as built: about 24,700 lines across the three services - 29 frontend pages, 33 frontend components, 19 REST controllers, 31 backend services, 27 JPA entities, 15 Flyway migrations, 12 ML pipeline modules, 37 git commits.

13 DATABASE DESIGN (PostgreSQL, 15 Flyway migrations)

Flyway applies numbered SQL files (V1 ... V15) at startup, so every machine has the same schema and the history is in git. Main tables:

ACCOUNTS AND ACCESS

users	account, role (ADMIN / COUNSELOR), status, profile fields
counselor_applications	what an applicant submitted, and what the admin judged
verification_documents	licence, degree, photo ID - typed, not an unlabelled pile
password_reset_tokens, email_verification_tokens	

CLINICAL WORKFLOW

participants	a person seen repeatedly, referenced as P-001 etc.
sessions	one recording: language, status, consent flags, paths, speaker attribution, failure reason, purge state
session_companions	other people present, with their enrolled voiceprints
session_features	the feature vectors (JSONB): text 3096, audio 85, video 111, plus the participant transcript
reports	prediction, confidence_score, per-modality contributions, and scoring_details (JSONB: cut-off, per-modality scores, fusion weights, out-of-range warnings, sentence attributions, face-region attributions)
explanation_factors	one row per explained feature, with its contribution
counselor_judgments	the counsellor's own assessment - the research label

SPEAKERS

counselor_voiceprints	256-dimensional embedding per counsellor
enrollment_passages	the admin-editable passage everyone reads

ML OPERATIONS

model_versions	every uploaded model: modality, language, label, file name, SHA-256, feature count, threshold, notes, who uploaded it, and exactly one active row per (modality, language)
----------------	--

GOVERNANCE

dataset_samples	research-dataset eligibility and admin review
audit_logs	every administrative action

Why feature vectors are JSONB and not columns: nothing queries an individual feature, and a retrained model with a different width must not require a schema migration.

14 BACKEND STRUCTURE (Spring Boot)

com.project.depression

controller/	19 REST controllers - Auth, Session, Participant, Profile, Settings, Voiceprint, ReferenceData, CounselorApplication, CounselorJudgment, Health, plus Admin{User, Counselor, Model, Dataset, Privacy, Stats, Audit, EnrollmentPassage}, and a GlobalExceptionHandler
service/	31 services - AuthService, SessionService, SessionProcessingService, ModelVersionService, VoiceprintService, DatasetExportService, MediaPurgeService, DataErasureService, AuditLogService, MailService, GoogleAuthService, CaptchaService, LanguageCatalogService, ...
entity/	27 JPA entities (section 13)
dto/	request/response records - the API never exposes entities directly
repository/	Spring Data JPA repositories
config/	SecurityConfig, JwtService, JwtAuthFilter, RateLimitFilter, UserDetailsServiceImpl, AdminAccountSeeder
client/	MLServiceClient - typed HTTP client for the ML service, with a 60-minute timeout for the processing call and 120 s elsewhere

Request flow: Controller (validates, authorises)
 -> Service (business rules, transactions)
 -> Repository (database)

Security: Spring Security + JWT. Every request carries a Bearer token; the filter validates the signature and loads the account; role checks are per route.

Session processing, specifically (SessionProcessingService):

set status PROCESSING in its OWN transaction (so the counsellor sees it immediately)
-> call the ML service -> store speaker attribution, transcripts, features, report, explanation factors -> status COMPLETED. A refusal from the ML service is mapped to TOO_SHORT / SPEAKER_UNVERIFIED / FAILED with the message preserved.

15 FRONTEND STRUCTURE (React 19 + Vite 8)

frontend/src

pages/	29 pages
public:	Login, RequestCounselorAccess, ForgotPassword, ResetPassword, VerifyEmail, PrivacyPolicy, TermsOfService, CrisisResources, HelpGuide, NotFound
counsellor:	Dashboard, Participants, ParticipantDetail, Sessions, NewSession, SessionDetail, Report, VoiceEnrollment, Settings, Profile
admin:	AdminHome, AdminUsers, AdminApplications, AdminApplicationDetail, AdminModels, AdminDataset, AdminDatasetDetail, AdminPrivacy, AdminVoicePassage
components/	33 components - ConfidenceRing, ModalityContributions, ExplanationFactorBar, PlainExplanation, SentenceAttributions, FaceRegionMap, JudgmentPanel, ConsentRecord, SpeakerAttributionPanel, SessionNotes, VoiceRecorder, PasswordInput, GoogleSignInButton, LoadingState / EmptyState / ErrorState, ...
lib/	featureGlossary.js - turns model feature names into plain English
api/	axios clients; the API base URL is derived from window.location so the same build works on localhost and over the LAN
index.css	one shared stylesheet: CSS custom properties (violet accent, warm neutral background, red reserved for genuine errors), utility classes, responsive breakpoints, WCAG AA contrast, visible keyboard focus

Design decisions worth defending in a viva:

- One stylesheet with design tokens, not a component library: the visual identity is deliberate ("clinical calm"), and every page inherits it automatically.
- Reusable Loading / Empty / Error states so no screen invents its own behaviour.
- The report is print-friendly: "Print / Save as PDF" uses the browser's own dialog, so there is no PDF dependency to maintain.

16 ML SERVICE STRUCTURE (FastAPI)

ml-service/app

main.py	/process, /enroll-voice, /health, /languages and the internal admin endpoints; maps refusals to clean HTTP codes
schemas.py	the request/response contract (Pydantic)
internal_admin.py	/internal/validate-model, /internal/reload-models, /internal/active-models - protected by a shared secret with a constant-time comparison
mock_inference.py	a mock pipeline with the identical response shape, for development without models
real/	
media_pipeline.py	ffmpeg -> faster-whisper -> pyannote diarization -> participant isolation
voiceprint.py	256-d embeddings, cosine matching, role resolution
text_features.py	24 lexical + 3,072 mpnet features (and the per-sentence vectors used for sentence attribution)
audio_features.py	the 85-column legacy set and the 64-column clean set
openface_features.py	runs OpenFace 2.2 and computes the 224 AU/gaze features; also maps each feature to a face region
video_features.py	MediaPipe face geometry, 111 features (stored for research)
feature_space.py	which extractor produces which column, and how a model's own column list is turned into its input vector
model_loader.py	reads the active manifest, loads bundles, hot-reload
adequacy.py	the too-short guard and its measurements
real_inference.py	the orchestrator: scoring, SHAP, sentence and face attribution
daic_transcript.py	writes the conversation in DAIC-WOZ format for future training

MODEL MANAGEMENT (the "no developer needed" part)

1. Admin uploads a .joblib on Admin -> Models.
 2. The backend stores it, hashes it (SHA-256) and asks the ML service to VALIDATE it: is it a dict with a model and a threshold, does it expose predict_proba, and is every column it names something this platform actually measures? Validation is by COLUMN NAME, not by counting columns - two different 224-column feature sets look identical to a width check.
 3. A rejected file is deleted and never recorded.
 4. Activating a version writes models/active_manifest.json and triggers a hot reload; the cache is only swapped once every model loads successfully.
 5. "Revert to built-in" restores the shipped weights (text/audio/fusion) or removes the model entirely (video, early fusion - there is no built-in for those).
 6. Rules enforced: a three-input fusion model cannot be activated without an active video model, and a video model cannot be removed while one depends on it.
- SECURITY NOTE: a .joblib is a Python pickle and loading one runs code. Upload is ADMIN-only, the internal endpoint needs the shared secret, and the ML service must never be exposed to the internet.

17 RUNNING THE PROJECT ON LOCALHOST (step by step)

17.0 PREREQUISITES (one-time)

- Docker Desktop (for PostgreSQL)
- Java 17 and Maven wrapper (bundled: ./mvnw.cmd)
- Node.js 20+ (frontend)
- Python 3.11 virtual environment in ml-service/venv with requirements.txt installed
- ffmpeg: the SHARED build, unzipped into ml-service/tools/
(torchcodec loads FFmpeg's DLLs at runtime; a static-only build fails)
- OpenFace 2.2 (only if a model using OpenFace features is active):
unzip OpenFace_2.2.0_win_x64.zip into ml-service/tools/ so that
ml-service/tools/OpenFace_2.2.0_win_x64/FeatureExtraction.exe exists
- Hugging Face token (HF_TOKEN) with the terms accepted for
pyannote/speaker-diarization-community-1 and pyannote/segmentation-3.0
- backend/.env : ADMIN_* seed account, ML_SERVICE_URL, ML_ADMIN_TOKEN, JWT_SECRET,
CORS_ALLOWED_ORIGINS, optional MAIL_* / Google / reCAPTCHA keys
- ml-service/.env: USE_REAL_MODELS=true, HF_TOKEN, ML_ADMIN_TOKEN (must match backend)

17.1 START POSTGRESQL

```
docker compose up -d
Verify: docker ps      (expect depression-postgres, healthy, 0.0.0.0:5433->5432)
```

17.2 START THE ML SERVICE (terminal 1)

```
cd ml-service
./venv/Scripts/python.exe -m uvicorn app.main:app --host 127.0.0.1 --port 8000
Verify: curl http://127.0.0.1:8000/health -> {"status":"ok","real_models":true}
```

17.3 START THE BACKEND (terminal 2)

```
cd backend
./mvnw.cmd -o spring-boot:run
Flyway applies any new migrations at startup.
Verify: curl http://localhost:8080/api/health -> HTTP 200
```

17.4 START THE FRONTEND (terminal 3)

```
cd frontend
npm install      (first time only)
npm run dev -- --host
Open http://localhost:5173
```

17.5 FIRST-RUN CHECKS

- Log in as the seeded admin account (credentials come from backend/.env).
- Admin -> Models: confirm which models are serving.
- Counsellor account: enrol the counsellor's voice (record or upload) BEFORE uploading a session - without it the pipeline cannot tell whose voice to analyse and will refuse the session.
- Upload a session with at least ~3 minutes of participant speech; shorter recordings are refused by design.

17.6 SHUTTING DOWN

Ctrl-C in each terminal; docker compose down (add -v ONLY if you want to erase the database - the pgdata volume holds every account, session and report).

17.7 TROUBLESHOOTING

- "Docker API not reachable": Docker Desktop is not running.
- Port already in use: another copy is still running - find it with `netstat -ano | findstr :8080` and stop that process.
- Sessions stuck in PROCESSING: the ML service died mid-request; restart it and re-upload.
- Language not selectable: that language has no complete model set; install models.

18 EXPLAINABLE AI (XAI) IN THIS PROJECT

18.1 Why the models are linear on purpose

Every shipped model is linear (logistic regression, with PCA which is also linear). That is a deliberate trade: a non-linear model might score slightly higher, but its prediction could only be explained by approximation. With a linear chain, the exact contribution of every feature can be computed in closed form - no sampling, no surrogate model, no "explanation" that is itself a guess.

18.2 SHAP - for text and audio (and video too, in fact)

SHAP (SHapley Additive exPlanations) assigns each feature the share of the prediction it is responsible for, with the guarantee that the shares add up to the prediction. For a linear model the Shapley value has a closed form:

$$\text{shap}_i = w_{\text{original}_i} * (x_i - \text{mean}_i) / \text{std}_i$$

which is what our service computes, using the weights back-mapped through the PCA (section 9). Properties we verified on the GPU machine and re-verified on the website:

- additivity: $\text{sum}(\text{shap}) + \text{intercept} = \text{the model's log-odds}$, max error $3.553\text{e-}15$
- agreement with the shap library's LinearExplainer: max error $6.193\text{e-}16$

So when the report says "used more negative words: +0.31", that number is the exact amount that feature moved the log-odds, not an approximation. On the report, the SHAP values are drawn as WATERFALL charts ("How the score was built (SHAP)"):

- one overall waterfall: base value (an average participant) -> + text SHAP -> + audio SHAP -> + video SHAP -> this session's score, with the cut-off marked;
- one waterfall per modality (tabs: What was said / How it was said / Facial movement): the 8 strongest features, then "N other measurements" summed, then the modality total. Each bar says how far that measurement was from the training average (for example "2.1 SD above").

The base value is computed as (model log-odds - sum of all SHAP values), so the chart lands exactly on the score the model returned, by construction. NOTE for anyone re-running the check: shap's default masker subsamples the background to 100 rows and produces a spurious $\sim 1\text{e-}3$ disagreement; build the masker explicitly.

18.3 Sentence-level attribution - the text explanation a human can act on

A SHAP value for "mpnet_734" is useless to a counsellor. But the text features are POOLED sentence embeddings, so the text score can be split back across the participant's own sentences, exactly:

mean block	sentence s contributes	$w \cdot v_s / n$
weighted-mean block		$w \cdot (\text{weight}_s * v_s)$
max block	each dimension's contribution belongs to the sentence that achieved the maximum in that dimension	
std block	NOT linear in a single sentence - reported as an "unattributed remainder", never spread around to make the totals look complete	

Verified: sentence totals + remainder reconstruct the text SHAP total to $5.55\text{e-}15$. On the report this becomes "Which answers moved the result", quoting the participant's own sentences with a bar showing how strongly each pushed, and in which direction.

18.4 GRAD-CAM: why true Grad-CAM is not possible, and the heat map we draw instead

Grad-CAM produces a heat map by taking the gradient of the class score with respect to the activations of a CONVOLUTIONAL feature map and projecting it back onto the INPUT IMAGE. It needs a convolutional network that reads pixels, and spatial feature maps. The shipped model has neither:

- No convolutional network. The video path is StandardScaler -> PCA -> logistic regression over tabular numbers. The gradient of the output with respect to the input is just the model's constant coefficient vector - identical for every session, so a gradient "heat map" would carry no session-specific information.
- No spatial dimension in the model's input. The 224 video features are per-session STATISTICS (mean, std, percentiles, frame-to-frame change) of action units and gaze.
- DAIC-WOZ ships no raw video, so no CNN could be trained on it.

WHAT WOULD BE NEEDED for true Grad-CAM: a consented raw-video dataset, a CNN trained on face crops with enough participants to fit it honestly, and evidence that such a model beats the action-unit features at all (video has scored at or below chance on this corpus in three separate runs).

WHAT WE DELIVER: a GRAD-CAM-STYLE FACE HEAT MAP on the participant's own face, built from the model's exact SHAP values rather than from gradients.

1. OpenFace tracks 68 face landmarks in every frame (the -2Dfp output), alongside the muscle intensities (action units) and gaze the model already uses.
2. Each video feature belongs to one facial muscle (AU04 = brow lowerer, AU15 = lip corner depressor, AU45 = blink, ...) or to gaze. Summing a muscle's SHAP values gives that muscle's exact push on the score.
3. Only pushes TOWARD the reported outcome are drawn - the same idea as Grad-CAM's ReLU, which keeps only positive evidence for the class being explained.
4. Frame choice: in each third of the interview, the frame where the face most expressed what the model weighed: $\text{argmax over frames of } \sum \text{ over muscles } (\text{push}_m * z_{mf})$
where z_{mf} is how active muscle m is in frame f , standardised within the session.
5. Heat on a frame: $\sum \text{ over muscles } (\text{push}_m * (0.4 + 0.6 * \text{activity}_{mf}))$, placed as a Gaussian on the landmarks of that muscle (brow points for AU04, mouth corners for AU12/AU15, eye points for blink and gaze ...), width 0.16 x the distance between the eye corners. All frames share one colour scale, so they can be compared. Colours use the classic Grad-CAM "jet" map (blue weak -> red strong).
6. The three stills are cropped around the face (256 x 256 JPEG) and stored in the report, because the recording itself is deleted after processing.

It is labelled "Grad-CAM-style" on the report, with a sentence saying the model reads measurements, not pixels. Below it the report keeps the FACE-REGION ATTRIBUTION for the whole session, which also covers gaze, overall movement and tracking quality. PRIVACY: three small face crops are a new kind of stored data. They live only inside the report: erasure deletes them with it, and the research export never reads report details.

18.5 What the counsellor actually sees on a report

1. Depression score as a ring, with the decision cut-off marked on it. 2. A caution box when measurements fall far outside the training range, naming the

worst one in plain words (for example "Speaking volume").

3. Modality contributions - text / audio / video shares, with a sentence explaining

that for a single fused model these are shares of the reasoning, not separate verdicts.

4. "How the score was built (SHAP)" - the overall waterfall and one waterfall each for

text, audio and video (section 18.2).

5. Explainability breakdown - the strongest features, each with what was observed and

which way it pushed the result, translated out of model jargon by a glossary.

6. "Which answers moved the result" - the participant's own sentences. 7. "What the face contributed" - three Grad-CAM-style heat maps on the participant's

face (section 18.4), then the shaded face diagram and region list.

8. "In plain words" - the same evidence written to be read aloud to the participant,

with an explicit statement that this is not a diagnosis and cannot decide anything on its own.

9. The counsellor's own assessment, stored separately and never fed back to the model.

19 RESULTS

19.1 Per-modality models, held-out official test split (47 people, 14 depressed)

Model	Algorithm	Inputs	Accuracy	F1	ROC-AUC	95% CI
Text	LogReg, SelectKBest 50, C=0.01	3096	70.2%	0.563	0.7532	[0.59-0.89]
Audio	LogReg, SelectKBest 25, C=1.0	64	57.4%	0.444	0.6017	[0.40-0.79]
Video	LogReg, SelectKBest 25, C=1.0	224	57.4%	0.474	0.5844	[0.41-0.75]

Reading: text carries most of the signal; audio is weak; video is barely above chance on its own. This is consistent with three independent runs on this corpus.

19.2 Fusion models compared (official protocol: train+dev 141 -> test 47, scored once)

Model	ROC-AUC	Accuracy	F1	Sens	Spec
Previous baseline (inherited work)	0.768	0.702	0.611	-	-
v1 late fusion (3 probabilities)	0.7684	0.7021	0.6111	0.786	0.667
v1 early fusion (concatenated, enet)	0.8333	0.766	0.667	0.786	0.758
v2 early fusion UNCONSTRAINED	0.8766	0.6809	0.6341	0.929	0.576
v2 early fusion CONSTRAINED (SHIPPED)	0.8463	0.766	0.6857	0.857	0.727

Best overall on the comparable protocol: the shipped v2 model - AUC 0.8463 vs the 0.768 baseline, F1 0.6857 vs 0.611, accuracy 0.766 vs 0.702, and it finds 12 of the 14 depressed participants.

19.3 The more reliable estimate: 10 repeated 80/20 splits, selection redone inside each

Configuration	inner-CV AUC (mean +/- std)
v1 text only (reference)	0.8255 +/- 0.032
v1 early fusion B1 (text+audio+video)	0.8140 +/- 0.033
v2 unconstrained	0.7633
v2 constrained (shipped)	0.7056

Single-split numbers are unstable at this sample size: one seed gave inner-CV 0.8796 but test AUC 0.5421 on its own 38 held-out people. That is exactly why the repeated protocol exists, and why we quote the repeated mean rather than a best split.

19.4 What each modality actually contributes in the shipped model

Share of the decision (back-mapped coefficient mass): text 77.1% audio 12.8% video 10.2%

Drop-one-modality, cross-validated AUC:

without text	0.4863	(-0.2193)	<- the model collapses to chance
without audio	0.7247	(+0.0191)	
without video	0.7569	(+0.0512)	

Honest reading: text carries the signal. Audio and video are included because the project requires a multimodal decision, and their inclusion costs a little accuracy on this corpus. We report the cost rather than implying they help.

19.5 The cost of forcing multimodality (v2)

cross-validated AUC	0.7633 -> 0.7056	(-0.0577)
seed-42 test AUC	0.8249 -> 0.7946	(-0.0303)
shares	text 96.1 / audio 0.9 / video 3.0	-> 77.1 / 12.8 / 10.2

19.6 Verification that the website computes the same thing as the training machine

- The website's text extractor reproduces the training features on the corpus:
24 lexical features exact to 3.55e-15; sentence vectors within 1e-3 (GPU half precision in v1; v2 fixed this by disabling TF32, giving parity at 6.4e-08).
- Participant timing features match to 1e-13 across all 188 participants.
- Running the training models on the corpus through the website's own code reproduces the published AUCs exactly: text 0.7532, audio 0.6017.
- Early-fusion attribution on the website: contributions + intercept equal the model's own log-odds to better than 1e-8 (checked again after v2's block weights were added).
- Model bundles delivered are byte-identical (SHA-256) to what was installed.

19.7 A bug the corpus let us find in our own website

DAIC-WOZ ships gaze in BOTH camera and head coordinates plus the head pose, so the website's conversion could be tested against ground truth. It was wrong: the code applied $R^{\wedge}T \cdot g$ where the corpus's own convention is $R \cdot g$. Mean error 0.2304 vs 0.0114 - a factor of 20. Fixed, and the fix is verified.

19.8 Comparison with published work (verified sources only)

Paper (year, venue)	Dataset	Fusion	Reported
Gratch et al. (2014, LREC)	DAIC	-	corpus paper
Valstar et al. (2016, AVEC/ACM)	DAIC-WOZ	challenge	RMSE 5.73, MAE 4.74
Al Hanai et al. (2018, Interspeech)	DAIC-WOZ	LSTM sequence	"comparable to topic modelling"
Rohanian et al. (2019, Interspeech)	DAIC-WOZ	word-level gated	claims to beat early and late fusion
Burdisso et al. (2024, ClinicalNLP/NAACL)	DAIC-WOZ	text only	F1 0.90 by EXPLOITING interviewer bias
Sun et al. (2021, Sensors/MDPI)	E-DAIC	LATE fusion	CCC 0.733, RMSE 3.783
Ishikawa & Duke (2026, preprint)	E-DAIC	text+LLM hybrid	macro-F1 0.723 (LOSO)
Gan et al. (2025, preprint)	DAIC-WOZ	teacher-student	F1 ~0.99 claimed - not credible
OURS (v2, shipped)	DAIC-WOZ	EARLY fusion	AUC 0.8463, F1 0.6857, acc 0.766

Two honesty notes for the viva:

- These numbers are NOT directly comparable: protocols, splits and even datasets differ (E-DAIC is a different, larger corpus; LOSO is a different validation).
- The headline 0.90 and 0.99 figures in the literature come from interviewer leakage or unverifiable protocols. Our contribution is a defensible number under a stated protocol, not a record.

20 STAKEHOLDERS: WHO DOES WHAT

PARTICIPANT (the person being screened)

Gives informed consent (recording, automated analysis, storage, research reuse - each a separate choice). Speaks in the interview. Has the right to have their data erased. Never logs in. The report is written so it can be read aloud to them.

COUNSELLOR (the primary user)

Applies for access with professional credentials; enrolls their own voice once; creates participants and sessions; uploads or records the conversation; reads the report; RECORDS THEIR OWN PROFESSIONAL ASSESSMENT, which is stored separately from the AI result and is the label used for future research; writes private notes. The counsellor - not the model - is the decision maker.

COMPANION (family member or friend present in the room)

Optionally enrolls their voice at the start of a session so the system can exclude their speech from the analysis. This is what stops a parent's voice being scored as the participant's.

ADMINISTRATOR

Approves or rejects counsellor applications after checking documents; manages model versions (upload, validate, activate, revert); edits the enrolment passage; reviews which consented sessions enter the research dataset; exports the dataset; performs erasure requests; reads the audit log. Admin is a fully trusted role by design - uploading a model runs code on the ML server.

RESEARCH / ENGINEERING TEAM (us)

Trains models on the GPU machine, verifies parity with the website, packages bundles, and is responsible for the honesty of every reported number.

GUIDE AND EVALUATION PANEL

Assess the work. Everything in this document is reproducible from the repository and the results folders.

(Not a stakeholder, but stated for completeness: the platform makes no autonomous decision about anyone. There is no automated referral, no alerting, and no action taken on a score.)

21 LIMITATIONS (stated plainly - a mentor will ask)

- L1 It is a SCREENING AID, not a diagnosis. In testing it is wrong about 3 times in 10.
- L2 Small data. 188 participants, a 47-person test set. A difference of a few points between models is inside the noise; our confidence intervals span 0.70-0.95.
- L3 Population mismatch. Training data is US adults interviewed by a virtual agent in English. Indian participants, accents, code-switching, and human interviewers are untested. Other languages are disabled until their own models exist.
- L4 Audio and video contribute little on this corpus. Forcing them to count (v2) costs accuracy. We ship it that way for multimodality, and say so.
- L5 Recording conditions change the measurements. Microphone level, camera, lighting and interview length can push features far outside the training range - we detect and display this, but we cannot correct it yet.
- L6 Train/deploy differences remain. DAIC-WOZ used an older face tracker (CLNF) and we run OpenFace 2.2; the corpus is segmented into utterance rows while our diarization produces speaker turns (we trained on turn-merged data to match, but it is not identical).
- L7 The PHQ-8 label is a questionnaire outcome, not a clinical diagnosis.
- L8 Short interviews are refused rather than scored. That is safe, but it does mean the system cannot help with brief conversations.
- L9 No automated test suite yet: verification is scripted parity checks, API tests and manual testing.

L10 Not yet deployed beyond localhost; no HTTPS hosting, and the ML service must be

kept off the public internet.

L11 Single-language models today (English). Marathi/Hindi require their own training.

22 ADVANTAGES / STRENGTHS

- A1 Honest evaluation end to end: official splits, one-shot test, repeated splits with selection redone inside each, bootstrap intervals, and published costs of every design choice.
- A2 Leak-free modelling: participant speech only, interviewer-dependent features removed, gender excluded - against a corpus where the published state of the art is partly an artefact of leakage.
- A3 Exact explainability, not approximation: SHAP in closed form, verified to $1e-15$, down to individual sentences and face regions.
- A4 Refusals instead of silent failure: too-short, unidentifiable speaker, missing face, and out-of-range warnings on the report.
- A5 Operable without a developer: models are uploaded, validated by column name, activated and hot-reloaded from the admin screen.
- A6 Real speaker handling: voiceprint enrolment and diarization mean the right person is analysed, and the evidence for that decision is stored with the session.
- A7 Privacy by design: consent per action, recordings purged after feature extraction, erasure, audit trail, and features stored so future retraining never needs the original video.
- A8 A report written for two audiences: a clinician's breakdown and a participant's plain-language explanation on the same page.
- A9 Reproducible: every model bundle carries its own column list, threshold, training statistics and metrics; the website validates all of it on upload.

23 FUTURE SCOPE

- F1 Collect consented, counsellor-labelled sessions through the platform and retrain on them. This is the single most valuable next step: it fixes the population mismatch (L3), the recording-condition mismatch (L5) and the train/deploy gap (L6) at once.
- F2 Calibrate for recording conditions - loudness normalisation, and per-device baselines - so a different microphone cannot move the score.
- F3 Indian-language models (Marathi, Hindi), which the architecture already supports: install a language's models and it becomes selectable.
- F4 Longitudinal tracking: the same participant across sessions, showing change over time rather than a single snapshot - the data model already supports repeat sessions.
- F5 If a consented raw-video dataset becomes available, train a CNN on face crops and then - and only then - genuine Grad-CAM becomes possible (section 18.4).
- F6 Automated test suite (JUnit, pytest, Vitest) with CI on every commit.
- F7 Secure deployment: HTTPS, the ML service on a private network, backups and retention automation.
- F8 Clinical validation study with a qualified clinician's diagnosis as the reference standard, rather than PHQ-8.
- F9 Fairness auditing across gender, age and accent once enough local data exists.

F10 Counsellor-facing analytics: cohort trends, and model-versus-counsellor agreement

over time, which is also how model drift would be detected.

24 QUESTION BANK - what a mentor or panel is likely to ask

THE PROJECT

Q1. Explain your project in one minute.

A counsellor uploads a recording of a counselling session. Our ML service separates the participant's voice from everyone else's using voiceprints and diarization, transcribes the speech with Whisper, and measures three things: what was said (word habits and sentence meaning), how it was said (pitch, loudness, timing) and how the face moved (OpenFace action units and gaze). One early-fusion model trained on the official DAIC-WOZ corpus scores all 3,384 measurements together and produces a depression score against a stated cut-off. Every report explains itself with exact SHAP values, down to which sentences and which parts of the face moved the result. On the official test split it reaches AUC 0.846, accuracy 76.6%, and finds 12 of 14 depressed participants.

Q2. What is novel about it compared with existing DAIC-WOZ work?

Most published work stops at a notebook. We built a deployable, secured platform with real speaker identification, leak-free modelling, exact explanations at sentence and face-region level, admin-managed model versioning with name-based validation, refusal guards, and a consent-governed research dataset labelled by the counsellor rather than by our own model.

Q3. Who are the users?

Admin and counsellor log in; participants and companions are recorded with consent. Section 20 lists each role's responsibilities.

Q4. Is this a medical device / can it diagnose?

No. It is a screening aid. The report states that, gives the error rate, and stores the counsellor's own assessment separately. No action is taken automatically.

DATASET

Q5. Which dataset, and why that one?

DAIC-WOZ (USC). It is the benchmark for depression detection from clinical interviews, used by the AVEC 2016/2017 challenges, and it has audio, transcripts and facial features with PHQ-8 labels for all participants.

Q6. How large is it, and how is it split?

189 in the split files; 188 usable after excluding participant 440 (corrupt in the release). Official split: 107 train, 34 dev, 47 test, participant-disjoint. 55 depressed (29.3%) and 133 not depressed.

Q7. How is the data labelled?

PHQ8_Binary: the 8-item Patient Health Questionnaire scored 0-24 and thresholded at ≥ 10 , the standard cut-off for probable depression.

Q8. Is the dataset balanced? How did you handle imbalance?

No - about 30% depressed. We did not oversample. We report AUC, F1, sensitivity and specificity rather than accuracy alone, and we choose the decision threshold by Youden's J instead of 0.5.

Q9. Did you find problems in the dataset?

Yes: participant 440 is corrupt; four participants are excluded by the release itself (342, 394, 398, 460); one participant's binary label disagrees with the score rule; three transcripts contain no interviewer rows; and the training split has an 18-point gender gap in depression rate that does not exist in dev/test.

Q10. Why not use the interviewer's questions - they are part of the conversation?

Because the interview branches on the answers, so the questions leak the outcome. A 2024 ACL-published study reached 0.90 F1 using ONLY the interviewer's prompts. That signal will not exist in a real counselling session, so

using it would inflate our numbers and fail in deployment.

Q11. Why did you use 80/20 when the corpus has an official split?

We use both. The 80/20 split, repeated 10 times, is our primary estimate because it uses more data for validation and is far more stable. The official split is our secondary evaluation, because it is the only number comparable with published work.

FEATURES

Q12. What are the 3,096 text features?

24 hand-counted lexical features (pronoun rates, absolutist words, negative and positive words, negations, hedges, past tense, sleep/fatigue words, answer lengths, vocabulary richness, ratios) plus 3,072 sentence-meaning features: each answer embedded by all-mpnet-base-v2 into 768 numbers, then pooled four ways - mean, length-weighted mean, max and standard deviation.

Q13. Why four poolings instead of just the mean?

Each captures something different. The mean is the overall tone; the weighted mean lets longer answers count more; the max preserves the single most extreme statement, which an average would erase; the standard deviation measures how much the meaning varied across the interview.

Q14. What are the 64 audio features, and why not the original 85?

Pitch (YIN, in semitones), loudness in dB, voiced ratio, jitter and shimmer proxies, answer durations, words per answer and speaking rate, each summarised by nine statistics. The original 85 included 24 features that depended on the interviewer's timing - response latency and between-turn gaps - which measure Ellie's scripted pacing rather than the participant. Removing them cost accuracy but is honest.

Q15. What are the 224 video features?

14 facial action-unit intensities and 6 presence rates from OpenFace, their total, and 12 gaze directions - each summarised by mean, std, percentiles, max and frame-to-frame movement. Action units are muscle movements: AU04 is frowning, AU12 smiling, AU45 blinking.

Q16. Why OpenFace and not a deep video model?

DAIC-WOZ ships no raw video - only pre-extracted facial features - so there is nothing to train a deep video model on. OpenFace produces the same family of measurements as the corpus, which is what makes training and deployment match.

Q17. How do you ensure the features at deployment match the features at training?

Three ways. The training run imports the website's own extraction functions; the bundle carries the exact list of column names, and the website builds each model's input BY NAME; and we verified on the corpus that the website reproduces the training features (lexical exact, timing to 1e-13) and reproduces the published AUCs exactly.

MODELS AND TRAINING

Q18. Which algorithm did you finally use?

One early-fusion pipeline: per-modality median imputation, standardisation and PCA, block weighting, then L2-regularised logistic regression over the 40 components.

Q19. Why logistic regression and not deep learning?

With 107-142 training people, deep models overfit. Earlier work on this corpus tried six deep and attention architectures - all lost to simple models. Linear models are also the reason our explanations can be exact rather than approximate.

Q20. What is PCA doing here?

Reducing each modality to a small number of uncorrelated components (text 20, audio 10, video 10). It prevents the 3,096 text columns from swamping 64 audio columns, and it keeps the total number of parameters small enough for 150 training people.

Q21. What is early fusion, in your own words?

Joining all the measurements from all modalities into one vector and training ONE model on it, so the model can use relationships between modalities. Late fusion instead trains one model per modality and combines their output probabilities.

Q22. Why is naive early fusion a bad idea?

Because of scale and width imbalance. We measured it: a global feature selection over all 3,384 columns chose 50 text columns and zero audio or video columns, so the "fusion" model returned byte-identical results to a text-only model.

Q23. How did you fix that?

Per-block PCA so each modality contributes a comparable number of components, per-block weights (text 0.2236, audio 0.6325, video 0.3162) so no block wins by width, and an L2 penalty so no block can be deleted.

Q24. Why did v1 give audio zero weight?

v1 used an elastic-net penalty, which contains L1 and can drive coefficients to exactly zero. It deleted the entire audio block. v2 uses L2 only, which shrinks but never deletes.

Q25. So does audio actually help?

On this corpus, no. Dropping audio changes cross-validated AUC by +0.019 and dropping video by +0.051 - both slightly better without them. We include them because the project requires a multimodal decision, and we report the cost: -0.0577 AUC.

Q26. How is the decision threshold chosen?

Youden's J - the threshold maximising sensitivity + specificity - 1 - computed on out-of-fold predictions of the training data only, never on the test set.

Q27. Why is the cut-off below 50%?

Because only about 30% of the training participants are depressed, so the model's probabilities sit low. At 0.5 it would miss most depressed people.

Q28. What is out-of-fold prediction and why does it matter?

A prediction made for a data point by a model that did not train on it. The late-fusion combiner is trained on out-of-fold probabilities so it never sees over-confident predictions, which would teach it the wrong weights.

Q29. How did you prevent overfitting?

Strong regularisation (C=0.1, L2), dimensionality reduction, simple models, selection redone inside every cross-validation split, a test set touched once, and confidence intervals on everything.

Q30. How do you know your best model is not just a lucky split?

Because we repeated the entire procedure on 10 different splits. One seed showed inner-CV 0.8796 but test AUC 0.5421 - proof that single-split numbers cannot be trusted at this sample size.

RESULTS

Q31. What accuracy do you get?

On the official test split, the shipped model: accuracy 76.6%, AUC 0.8463, F1 0.6857, sensitivity 85.7%, specificity 72.7%. Repeated-split cross-validated AUC 0.7056 for the constrained model (0.7633 unconstrained).

Q32. Is that better than what you started with?

Yes. The inherited baseline on the same protocol was AUC 0.768, F1 0.611, accuracy 0.702. We improved all three.

Q33. Why is your accuracy lower than papers claiming 90-99%?

Because those numbers mostly come from interviewer leakage, tuning on the test set, or unverifiable protocols. The one paper that reached 0.90 F1 did so by deliberately exploiting the interviewer bias - it was a warning, not a result to copy.

Q34. Which modality is most useful?

Text, by a wide margin: 77% of the decision weight, and removing it collapses the model to chance (AUC 0.486).

Q35. What does ROC-AUC mean?

The probability that the model scores a randomly chosen depressed person above a randomly chosen non-depressed person. 0.5 is guessing, 1.0 is perfect. It is independent of the threshold and of class balance.

Q36. Why report F1 and balanced accuracy rather than accuracy alone?

Because 70% of the corpus is not depressed, so a model that always said "not depressed" would be 70% accurate and useless.

Q37. Your confidence intervals are wide. Why report them?

Because with 47 test people they honestly are wide (roughly 0.72-0.95 for our AUC). Hiding that would imply a precision we do not have.

EXPLAINABLE AI

Q38. What is SHAP?

A method from cooperative game theory that splits a prediction among its features so that the parts add up to the whole. For a linear model it has a closed form: $\text{weight} \times (\text{value} - \text{training mean}) / \text{training std.}$

Q39. Did you use the shap library?

We compute the closed form directly - it is exact and needs no background sampling - and we cross-checked it against shap's LinearExplainer: agreement to $6.19\text{e-}16$. We also verified additivity (contributions + intercept = the model's log-odds) to $3.55\text{e-}15$.

Q40. How do you explain 3,072 embedding numbers to a counsellor?

Two ways. They are summed into one item, "what was said, overall", which is exact for a linear model. And the text score is split across the participant's OWN sentences - exactly for three of the four pooling blocks, with the fourth reported as an unattributed remainder rather than being faked.

Q41. Why not true Grad-CAM for the video part?

Grad-CAM needs a convolutional network that reads pixels, and its feature maps. Our video path is a linear model over 224 per-session statistics, and DAIC-WOZ ships no raw video to train a CNN on. Its gradient would be the model's constant coefficients - identical for every session, carrying no session-specific information.

Q42. So what do you show for video instead?

A Grad-CAM-style heat map on three frames of the participant's own face. Each facial muscle's exact SHAP value is placed on the landmarks OpenFace tracked for that muscle, scaled by how active the muscle was in that frame, keeping only pushes toward the reported outcome (like Grad-CAM's ReLU). The frames are the moments where the face most expressed what the model weighed. Under it, a face-region attribution for the whole session. It is labelled Grad-CAM-style, never Grad-CAM, because the heat comes from SHAP values of measurements rather than gradients over pixels.

Q43. What would make Grad-CAM possible later?

A consented raw-video dataset, a CNN trained on face crops with enough participants, and evidence that such a model beats the action-unit features at all.

Q44. Can the counsellor see WHY a particular person was flagged?

Yes: the strongest features with direction, the participant's own sentences that moved the score, the face regions that contributed, the modality shares, and a plain-language summary written to be read aloud.

SYSTEM AND ENGINEERING

Q45. Explain your architecture.

Three tiers: React frontend; Spring Boot backend with PostgreSQL for accounts, sessions and reports; a FastAPI ML service for the pipeline and models. The frontend talks only to the backend with JWT; the backend calls the ML service over an internal API protected by a shared secret.

Q46. Why separate the ML service?

Language fit, independent restart and scaling, and security: it is the process that loads .joblib files, and loading one executes code.

Q47. How does a model get updated without redeploying?

The admin uploads a .joblib; the backend hashes and stores it and asks the ML service to validate it by column NAME; on activation a manifest is written and the ML service hot-reloads. The cache is swapped only after every model loads successfully.

Q48. How is the model upload secured?

ADMIN role only, JWT-authenticated, the internal endpoint needs a shared secret compared in constant time, every file is SHA-256 hashed and audit-logged, size-capped at 520 MB, and the page warns that uploading a model runs its code.

Q49. What database design did you use and why Flyway?

27 entities across accounts, clinical workflow, speakers, ML operations and governance. Flyway gives versioned, repeatable migrations (V1-V15), so every machine has the same schema and the change history lives in git.

Q50. How do you identify the participant in the recording?

The counsellor (and any companion) enrol by reading a fixed passage, producing a 256-dimensional voice embedding. Diarized speakers are matched by cosine similarity (≥ 0.55 and a 0.06 margin); whoever is unmatched is the participant. If it is ambiguous the session is refused.

Q51. What happens if the ML service is down?

The session is marked FAILED with a readable reason and can be retried; the rest of the application keeps working.

Q52. How did you test the system?

Parity tests against the corpus (features and AUCs reproduced exactly), API integration tests of the model upload and activation rules, end-to-end pipeline runs on real recordings, unit checks of the explanation mathematics, and manual testing of every page and role. There is no automated unit-test suite yet - that is future work.

Q53. Give an example of a real bug you found and fixed.

Four: (1) every short recording came back "depressed" because the text model returned a constant far outside its training range - fixed with the adequacy guard; (2) an explanation said "spoke loudly" when the value was below average, because direction was inferred from the contribution's sign - now read from the measurement; (3) the gaze conversion applied the rotation transposed, a 20x error caught against the corpus's own ground truth; (4) the report labelled the score "confidence", so "Depressed, 35%" looked contradictory - now labelled "depression score" with the cut-off drawn on the ring.

Q54. What privacy protections are in place?

Consent recorded per action; recordings purged after features are extracted; features and transcripts retained instead of media; per-participant erasure; audit logging; role-based access; and the research dataset is opt-in and labelled by the counsellor.

Q55. What is stored after a session, and why?

The feature vectors, the participant transcript, the report and its explanation, and the speaker-attribution evidence. That is exactly what a future retraining needs, and it is why the original video can be deleted safely.

BIGGER PICTURE

Q56. What is the biggest weakness of this project?

That the signal is almost entirely in the text, so the multimodal claim is weaker than it looks on this corpus - and that our training population (US adults, virtual interviewer) does not match our intended users.

Q57. How would you improve accuracy next?

Not by another architecture on this corpus. By collecting consented, counsellor- labelled local sessions and retraining on them - that addresses population, recording conditions and segmentation together.

Q58. Can someone fake the result?

Yes, as with any interview-based screening. It is designed for cooperative counselling sessions, and the counsellor's judgement is final.

Q59. What are the ethical risks?

A false negative may delay help; a false positive may label someone wrongly. We mitigate by tuning for sensitivity, by stating the error rate on the report, by refusing unusable input, by never acting automatically, and by keeping the counsellor in charge.

Q60. Why should anyone trust your numbers?

Because every one of them is reproducible from the repository: the split files, the protocol, the model bundle with its own metrics, the verification logs showing the website reproduces the training features and AUCs, and the decisions log recording every deviation and its cost.

25 GLOSSARY

Action Unit (AU)	one facial muscle movement from the Facial Action Coding System (AU04 frown, AU12 smile, AU45 blink)
AUC / ROC-AUC	chance that a random depressed person scores above a random non-depressed one; 0.5 = guessing
Balanced accuracy	average of sensitivity and specificity
Bootstrap CI	resampling the test people many times to see how much a metric could vary
Diarization	deciding who spoke when in a recording
Early fusion	joining all modalities' features into one vector, one model
Embedding	a vector of numbers representing meaning (sentences) or identity (voices)
F1	harmonic mean of precision and recall for the depressed class
Imputation	filling a missing measurement (here, with the training median)
Late fusion	combining the output probabilities of per-modality models
Log-odds	the linear model's raw output z ; probability = $1/(1+e^{-z})$
OOF (out-of-fold)	a prediction made by a model that did not train on that point
PCA	principal component analysis - compresses many correlated columns into a few uncorrelated ones
PHQ-8	8-item depression questionnaire; ≥ 10 means probable depression
Precision	of those flagged depressed, the share who are

Recall / sensitivity of the depressed people, the share flagged

Regularisation	penalising large coefficients so a model generalises (L1 can zero them out; L2 only shrinks)
SHAP	a feature's exact share of a prediction, summing to the prediction
Specificity	of the non-depressed people, the share correctly cleared
Standardisation	converting to $(\text{value} - \text{mean}) / \text{standard deviation}$
Voiceprint	a 256-number summary of a voice, used to recognise a speaker
Youden's J	sensitivity + specificity - 1; maximised to choose the threshold

26 CURRENT STATUS OF THE DEPLOYMENT (as of 12 September 2026)

INSTALLED AND SERVING RIGHT NOW

- Full platform running locally: PostgreSQL (Docker, port 5433), Spring Boot backend (8080), FastAPI ML service (8000), React frontend (5173).
- SERVING MODEL: early fusion v2 (models/early_fusion_v2.joblib), uploaded and activated through Admin -> Models on 12 September 2026. 3,384 inputs, decision cut-off 0.4412, turn segmentation.
- Verified live on a real session end to end: the single model produced the score, and all three modalities contributed - text 70.3%, audio 19.2%, video 10.6% of the reasoning for that session (the model's trained shares are 77.1 / 12.8 / 10.2).
- The report for that session carried: the depression score with its cut-off marked, out-of-range warnings per modality, the participant's own sentences ranked by how much they moved the result, and a face-region attribution spread across eyes, brows, gaze, jaw and mouth.
- Also installed and versioned but not serving while v2 is active: the DAIC-WOZ text, audio, video and 3-way late-fusion models, and early fusion v1. Any of them can be restored from Admin -> Models without a redeploy.
- Verified: feature parity with training, exact attribution mathematics (contributions + intercept = the model's log-odds), model file integrity (SHA-256), and the corrected gaze conversion.

KNOWN AND DELIBERATE

- The report does not quote an error rate. It says the result is screening support, not a diagnosis, and that the counsellor's judgement is what counts; the measured accuracy is documented in section 19.
- Audio and video are included because the project requires a multimodal decision; section 19.4 states honestly what they contribute and what they cost.

REPOSITORY

- Public GitHub repository; the corpus, trained bundles, recordings and all secrets are excluded by .gitignore.
- 37 commits, every one describing what changed and why.